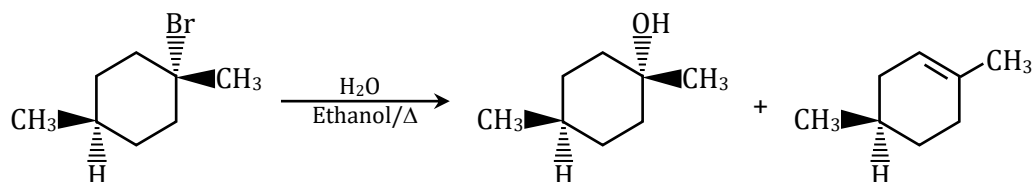




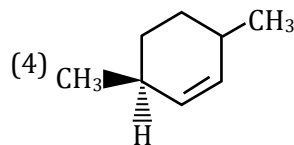
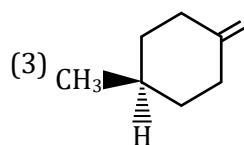
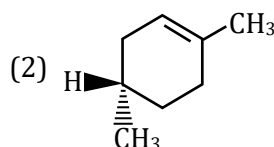
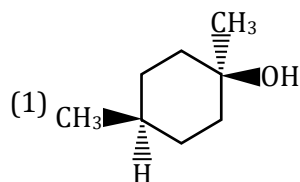
Alkyl Halides or Haloalkanes : RACE-01

Substitution Rx^n , Elimination Rx^n , S_N vs E , Nucleophilicity vs Basicity

1. Consider the following $E1/S_N1$ reaction



The missing product(s)



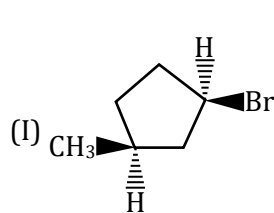
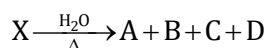
- (A) 1, 2 and 3 (B) 3 and 4 (C) 2 and 3 (D) 1, 2, 3 and 4

CHL158

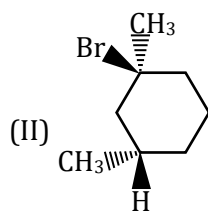
2. Select the reagent that will yield the greater amount of substitution on reaction with CH_3-CH_2-Br :
- (A) CH_3CH_2OK in dimethyl sulfoxide (DMSO)
 (B) $(CH_3)_3COK$ in dimethyl sulfoxide (DMSO)
 (C) Both (A) and (B) will give comparable amounts of substitution
 (D) Neither (A) nor (B) will give any amount of substitution

CHL159

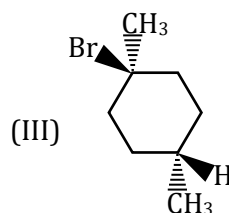
3. Under the specified conditions, substrate X undergoes substitution and elimination reactions to give products A –D. A and B are stereoisomers, but not enantiomers. C and D are enantiomers. A is not an isomer of C. Which of the following could be the starting material X?



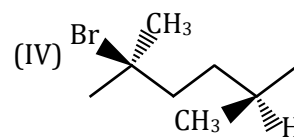
(A) (I)



(B) (II)



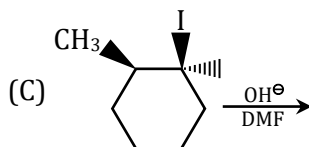
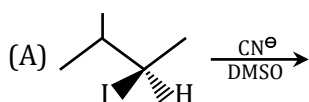
(C) (III)



(D) (IV)

CHL160

4. Which reaction results in the formation of pair of enantiomers?



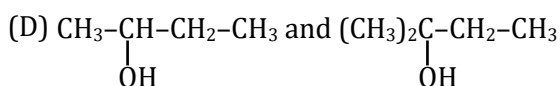
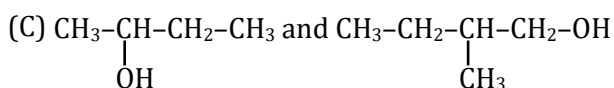
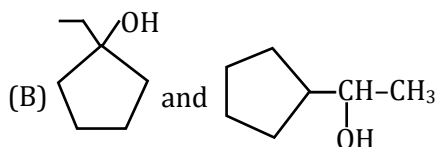
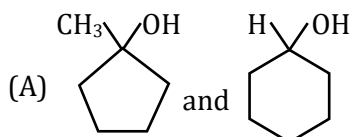
CHL161

5. Which of the following statement is true?

- (A) $\text{CH}_3\text{CH}_2\text{S}^\ominus$ is both a stronger base and more nucleophilic than $\text{CH}_3\text{CH}_2\text{O}^\ominus$
 (B) $\text{CH}_3\text{CH}_2\text{S}^\ominus$ is a stronger base but is less nucleophilic than $\text{CH}_3\text{CH}_2\text{O}^\ominus$
 (C) $\text{CH}_3\text{CH}_2\text{S}^\ominus$ is a weaker base but is more nucleophilic than $\text{CH}_3\text{CH}_2\text{O}^\ominus$
 (D) $\text{CH}_3\text{CH}_2\text{S}^\ominus$ is both a weaker base and less nucleophilic than $\text{CH}_3\text{CH}_2\text{O}^\ominus$

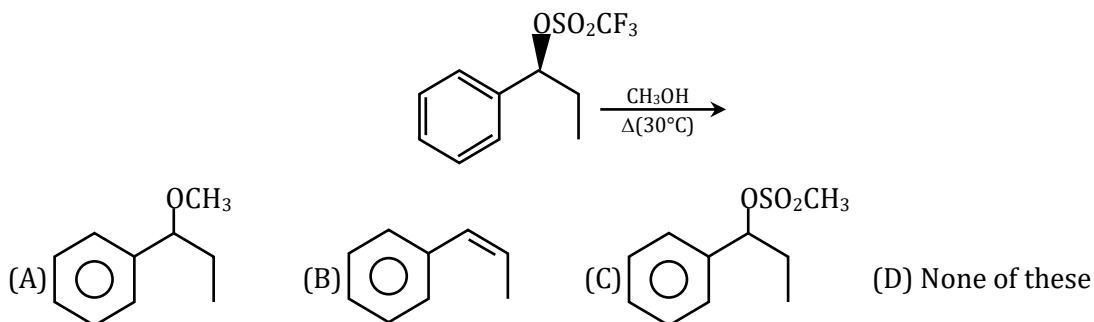
CHL162

6. In the given pair of alcohols, in which pair second alcohol is more reactive than first towards hydrogen bromide?



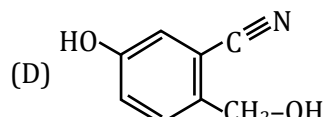
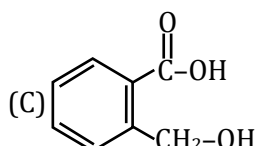
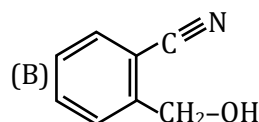
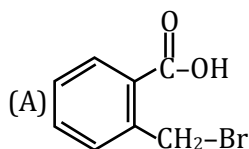
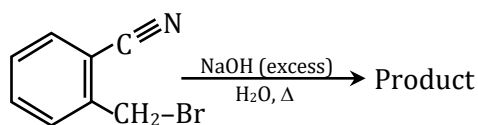
CHL163

7. Which product would be expected to predominate in the given reaction?



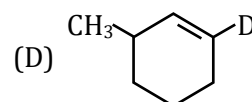
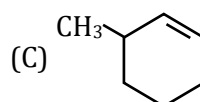
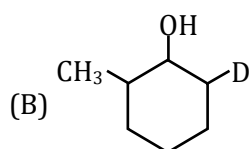
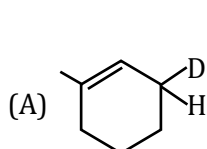
CHL164

8. Which is the major product of the following reaction?



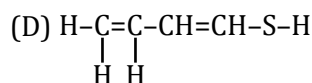
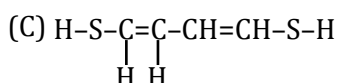
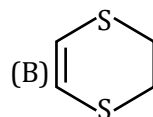
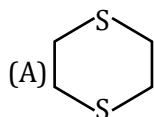
CHL165

9.
 $\xrightarrow{\text{Alc. KOH}}$ (A); Major product of this reaction is :



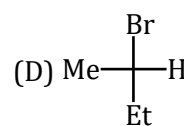
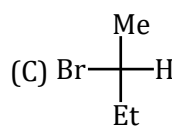
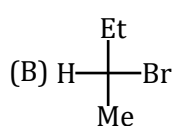
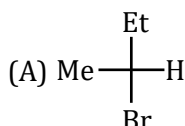
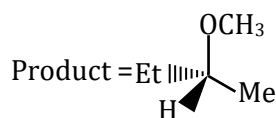
CHL199

10. 1,2-dichloro ethane + $\text{NaSCH}_2\text{CH}_2\text{SNa} \rightarrow (\text{P}) + 2\text{NaCl}$
Unknown product (P) of the above reaction is :



CHL167

11. The back-side attack on 2-bromobutane by methoxide (CH_3O^-) gives the product shown below. Which fisher projection represents 2-bromobutane used as the reactant in this reaction?

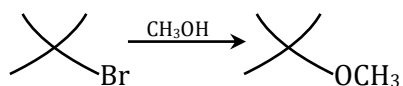


CHL168

12. Which alkyl halide has maximum density?
(A) $\text{C}_3\text{H}_7\text{I}$ (B) $\text{C}_2\text{H}_5\text{I}$ (C) CH_3I (D) CH_3Br

CHL169

13. Which of the following would be true for the reaction shown?



- (A) The rate of the reaction depends only on the alkyl bromide concentration.
(B) The rate of the reaction depends only on the methanol concentration.
(C) The rate of the reaction depends on both the alkyl halide concentration and the methanol concentration.
(D) The rate of the reaction depends on the concentration of neither reactant.

CHL200

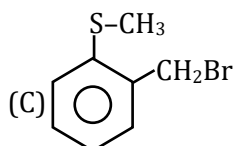
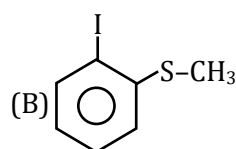
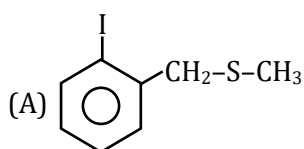
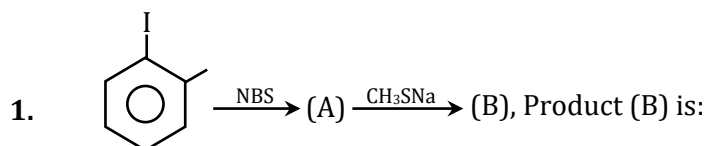
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	C	B	C	D	A	C	C	A
Question	11	12	13							
Answer	D	A	A							



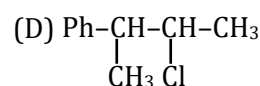
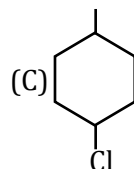
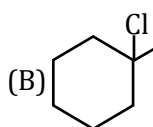
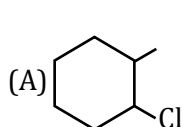
Alkyl Halides or Haloalkanes : RACE-02

Substitution Rx^n , Elimination Rx^n , S_N vs E , Nucleophilicity vs Basicity, Reaction of alkyl halide

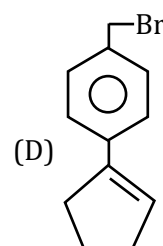
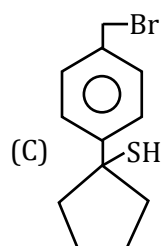
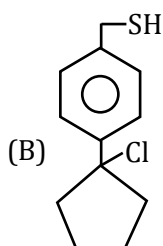
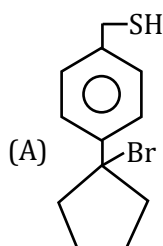
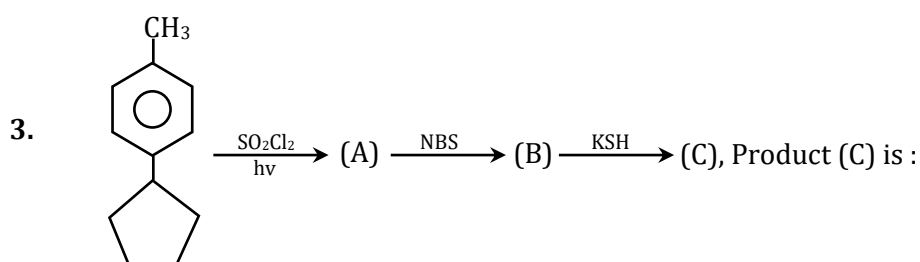


(D) None of these

CHL172

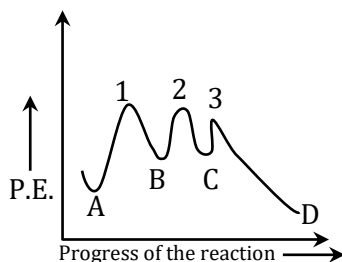


CHL173



CHL174

4. Energy profile diagram for an exothermic reaction, $A \xrightarrow{1} B \xrightarrow{2} C \xrightarrow{3} D$ is given below.



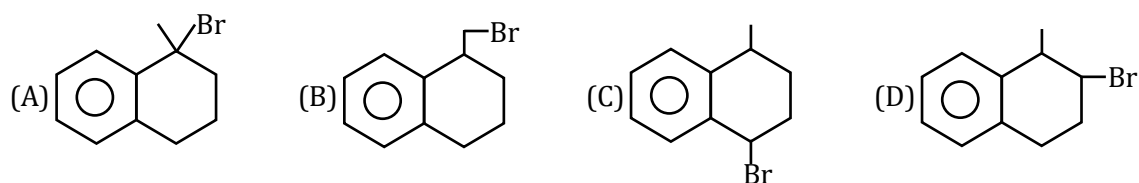
The rate determining step of the reaction is :

- (A) $A \longrightarrow B$ (B) $B \longrightarrow C$ (C) $C \longrightarrow D$ (D) can not predict

CHL175

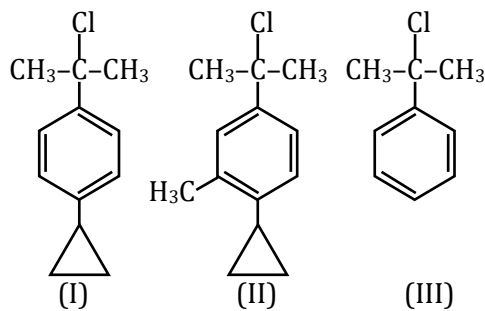
5. (A) major

Major product is (A) is :



CHL176

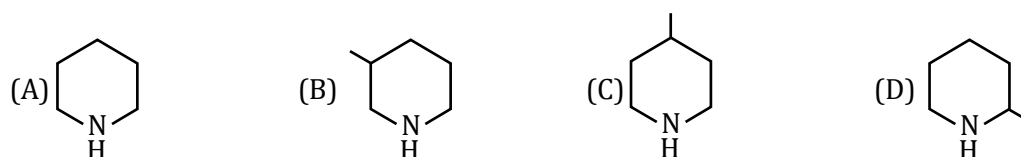
6. The decreasing order of reactivity of the compounds given below towards solvolysis under identical conditions is :



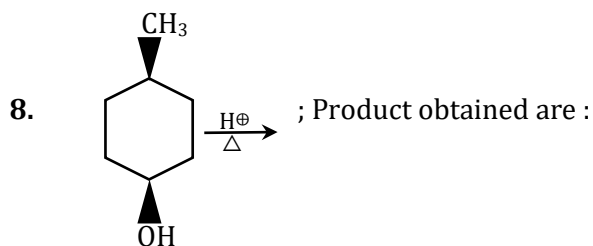
- (A) II > III > I (B) I > II > III (C) III > II > I (D) II > I > III

CHL177

7. (A) $\xrightarrow[(2) \text{ Ag}_2\text{O}/\text{H}_2\text{O}]{(1) \text{ excess CH}_3\text{I}/\text{K}_2\text{CO}_3}$ (B) $\xrightarrow[(iii) \Delta]{(1) \text{ excess CH}_3\text{I}/\text{K}_2\text{CO}_3, (2) \text{ Ag}_2\text{O}}$ $\text{H}_2\text{C}=\text{CH}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}=\text{CH}_2$ Identity A :



CHL179



(A) Racemic

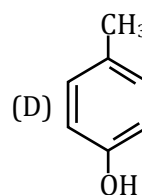
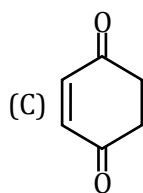
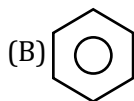
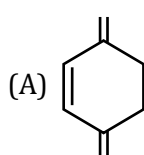
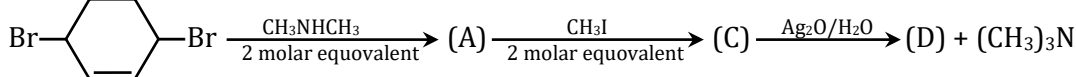
(B) Diastereomers

(C) G.I

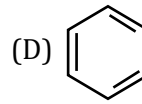
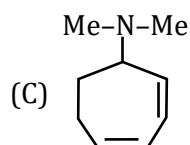
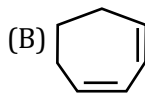
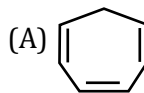
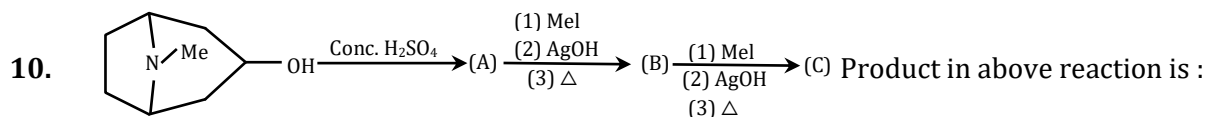
(D) Positional isomers

CHL180

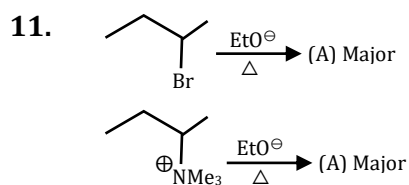
9. End product (D) in the given sequence is :



CHL181



CHL182



Relation between (A) and (B) is:

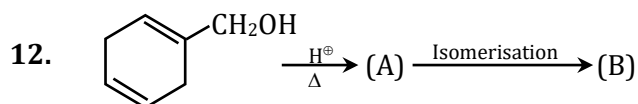
(A) G.I.

(B) Positional isomer

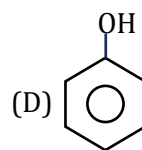
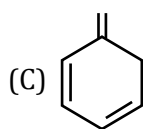
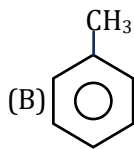
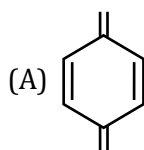
(C) Enantiomer

(D) Chain isomer

CHL183

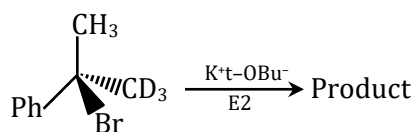


(A) on heating isomerizes to (B). What is the structure of (B) ?



CHL184

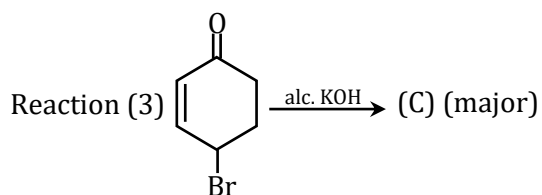
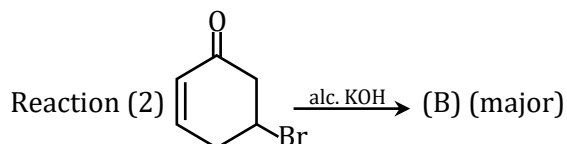
13. Which best describes the product of the following reaction?



- (A) Absolute configuration has been inverted
 (B) Absolute configuration has been retained
 (C) Racemization (loss of absolute configuration) has occurred
 (D) Loss of chirality has occurred (the product is achiral)

CHL185

14. Reaction (1) (A) (major)

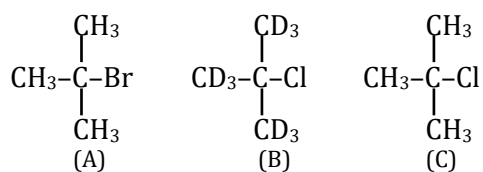


Product obtained in above reaction (1), (2) & (3) is :

- (A) A = B but C is different
 (B) A = C, but B is different
 (C) B = C, but A is different
 (D) A = B = C all products are identical

CHL186

15. Which of the following is true regarding increasing reactivity in the E2 reaction with $\text{Na}^+ \text{C}_2\text{H}_5\text{O}^-$?



- (A) A is more reactive than B
 (B) B is more reactive than A
 (C) A is more reactive than C
 (D) C is more reactive than B

CHL188

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	A	A	D	C	A	B	A
Question	11	12	13	14	15					
Answer	B	B	D	B	ACD					